Customer Engagement & Product Utilization Analytics for Retention Strategy

Research Paper | Customer Churn, Engagement, Product Adoption and Retention

Abstract

This study analyses 10,000 customer records to examine engagement, product utilization and customer churn. The observed churn rate is 20.37% (2,037 customers). Descriptive summaries and statistical association tests are used to identify customer groups for retention review. Findings are observational and do not establish causality.

1. Research Problem

Customer attrition can reduce customer lifetime value and weaken long-term banking relationships. This study investigates customer engagement and product-utilization patterns associated with churn.

2. Project Introduction

This project applies customer-level analytics to examine the relationship between activity status, product holdings, customer characteristics and exit status.

3. Business Background

Banks operate in a competitive environment where retaining existing customers is an important relationship-management objective. Activity and product holdings can help identify groups for service and engagement reviews.

4. Research Objectives

Measure the overall customer churn and retention baseline.

Compare churn between active and inactive customers.

Evaluate churn patterns by product count.

Examine geography, gender, age and tenure differences.

Identify inactive high-balance customers for further review.

Define retention KPIs and test selected statistical associations.

Develop practical retention recommendations.

5. Dataset and Variables

The analysis uses 10,000 usable records from European_Bank_Cleaned_Analytical.csv. The original dataset includes customer identification, credit score, geography, gender, age, tenure, balance, product count, credit card status, activity status, estimated salary and exit status. Exited is the target outcome (1 = exited, 0 = retained).

6. Data Preparation Methodology

The project uses a separate cleaned analytical dataset while preserving the raw source. This report loads the existing cleaned dataset, standardizes column names in memory, converts analytical fields to numeric types and excludes records missing essential analysis fields. It does not overwrite either source dataset.

7. Feature Engineering

Derived features include age bands and a high-balance segment based on the 75th percentile of Balance. Activity status and product count are used as engagement and utilization proxies, not direct measures of customer satisfaction.

8. Exploratory Data Analysis Findings

Usable records: 10,000.

Exited customers: 2,037.

Retained customers: 7,963.

Overall churn: 20.37%.

Overall retention: 79.63%.

Geographic Analysis

Customer distribution and churn by geography.

Geography	Customers	Exited	Churn_Rate
France	5014	810	16.1548
Germany	2509	814	32.4432
Spain	2477	413	16.6734

Gender Analysis

Customer distribution and churn by gender.

Gender	Customers	Exited	Churn_Rate
Female	4543	1139	25.0715
Male	5457	898	16.4559

Age Analysis

Customer distribution and churn by age group.

Age_Group	Customers	Exited	Churn_Rate
18-25	611	46	7.5286
26-35	3542	301	8.4980
36-45	3736	733	19.6199
46-55	1311	663	50.5721
56+	800	294	36.7500

Tenure Analysis

Customer distribution and churn by tenure.

Tenure	Customers	Exited	Churn_Rate
0	413	95	23.0024
1	1035	232	22.4155
2	1048	201	19.1794
3	1009	213	21.1100
4	989	203	20.5258
5	1012	209	20.6522
6	967	196	20.2689

Tenure	Customers	Exited	Churn_Rate
7	1028	177	17.2179
8	1025	197	19.2195
9	984	213	21.6463
10	490	101	20.6122

9. Engagement Findings

Active customers: 5,151; churn: 14.27%.

Inactive customers: 4,849; churn: 26.85%.

Engagement Retention Ratio (ERR): 1.1720.

Observed churn differs between active and inactive customers. This association does not prove causation.

10. Product Adoption Findings

Product count is treated as a proxy for product utilization. Customer counts and churn rates are shown below.

NumOfProducts	Customers	Exited	Churn_Rate_Percent
1	5084	1409	27.7144
2	4590	348	7.5817
3	266	220	82.7068
4	60	60	100.0000

11. Financial and Premium Customer Risk

The high-balance threshold is 127,644.24, representing the 75th percentile of Balance. The segment contains 2,500 customers; 1,247 are inactive. HBDR is 49.88%. Churn within inactive high-balance customers is 30.47%. This is a screening segment, not a verified premium-customer classification.

12. KPI Definitions and Results

ERR = active-customer retention divided by inactive-customer retention. PDI = average number of products held. HBDR = inactive customers among high-balance customers. CCS = cardholder retention minus non-cardholder retention, in percentage points.

KPI	Value	Unit
Total Customers	10,000.0000	Customers
Exited Customers	2,037.0000	Customers
Retained Customers	7,963.0000	Customers
Customer Churn Rate	20.3700	%
Customer Retention Rate	79.6300	%
Active Member Churn Rate	14.2691	%
Inactive Member Churn Rate	26.8509	%
Engagement Retention Ratio (ERR)	1.1720	Ratio
Product Diversity Index (PDI)	1.5302	Average products
High-Balance Dormancy Rate (HBDR)	49.8800	%

KPI	Value	Unit
Cardholder Retention	79.8157	%
Non-Cardholder Retention	79.1851	%
Cardholder Churn Spread (CCS)	0.6307	Percentage points

13. Statistical Validation

Chi-square tests evaluate associations between categorical variables and Exited; Cramer's V is reported as an effect size. Mann-Whitney U tests compare numeric-variable distributions between exited and retained groups. Tests are two-sided. A p-value below 0.05 indicates statistical evidence of association, not causation. Effect size and business relevance should also be considered.

Variable	Test	Statistic	P_Value	Effect_Size	Effect_Size_Name
Geography	Chi-square independence	301.2553	0.0000	0.1736	Cramer's V
Gender	Chi-square independence	112.9186	0.0000	0.1063	Cramer's V
IsActiveMember	Chi-square independence	242.9853	0.0000	0.1559	Cramer's V
NumOfProducts	Chi-square independence	1,503.6294	0.0000	0.3878	Cramer's V
HasCrCard	Chi-square independence	0.4713	0.4924	0.0069	Cramer's V
Age	Mann-Whitney U	11,874,649.5000	0.0000	Not available	Not calculated
CreditScore	Mann-Whitney U	7,839,548.0000	0.0199	Not available	Not calculated
Tenure	Mann-Whitney U	7,948,575.5000	0.1622	Not available	Not calculated
Balance	Mann-Whitney U	9,371,186.5000	0.0000	Not available	Not calculated
EstimatedSalary	Mann-Whitney U	8,250,768.0000	0.2271	Not available	Not calculated

14. Business Recommendations

Prioritize relationship reviews for inactive customers.

Investigate high-churn product-count groups and validate group sizes before intervention.

Review inactive high-balance customers through a service-led retention process.

Use suitability and customer-needs checks before cross-selling products.

Monitor churn, retention and engagement KPIs over time.

15. Limitations

The dataset is a customer snapshot and does not establish the timing or causes of exit.

Activity and product count are proxies; satisfaction and interaction history are not directly measured.

Observational associations do not prove causation.

The analysis does not measure the effectiveness of retention interventions.

The high-balance threshold is a statistical rule, not a formal premium-customer definition.

Results may not generalize to other banks, years or populations.

16. Conclusion

The observed churn rate is 20.37%. Activity status and product count show differences in churn across customer groups. These findings provide a baseline for targeted customer review and retention measurement. Future work should incorporate longitudinal behaviour and intervention outcomes.

17. References

European Bank customer dataset supplied for this project.

Pandas documentation: https://pandas.pydata.org/docs/.

SciPy documentation: https://docs.scipy.org/doc/scipy/.

ReportLab documentation: https://www.reportlab.com/docs/reportlab-userguide.pdf.

18. Reproducibility

Source file: C:\Users\hp\Customer Engagement & Product Utilization Analytics for Retention Strategy\data\processed\European_Bank_Cleaned_Analytical.csv. Report generated: 2026-11-03. Results are based on the records successfully loaded in this run.